



TUNIS BUSINESS SCHOOL
University of Tunis

GEOPOLITICS, OIL SHOCKS & MACROECONOMICS

Panel Data Analysis of the 2026 Gulf War Crisis

Econometrics — Academic Year 2025–2026

Countries Studied: USA · Italy · Saudi Arabia · Turkey

Panel: $N = 4$ countries · $T = 42$ months · $N_{total} = 168$ observations

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Methodology: Pooled OLS · Fixed Effects · Random Effects · Arellano Robust SE

1. INTRODUCTION & RESEARCH CONTEXT

The 2026 Gulf War crisis triggered an unprecedented geopolitical oil shock, disrupting global energy markets and transmitting inflationary pressures across interconnected economies. This project investigates the macroeconomic transmission mechanisms of this shock through a rigorous panel data econometric framework, examining how oil price volatility affects consumer price inflation (CPI) differently across oil-importing and oil-exporting nations.

Our panel comprises four economies — the United States, Italy, Saudi Arabia, and Turkey — chosen to capture structural diversity between advanced and emerging markets, and between oil importers and exporters. By leveraging 42 months of monthly data (January 2023 – June 2026), this study provides a “crisis snapshot” that isolates both the pre-crisis baseline and the post-March 2026 shock period.

Research Questions

- ▶ Does the oil price shock (measured as monthly % change in Brent crude) significantly affect CPI inflation across the panel?
- ▶ Is the exchange rate (NEER) a primary transmission channel for inflationary pressure?
- ▶ Do oil-importing nations face asymmetrically higher inflationary burdens compared to exporters?
- ▶ Which panel estimator — Pooled OLS, Fixed Effects, or Random Effects — best captures the underlying data-generating process?

Key Variables

Variable	Name in Model	Description
Dependent (CPI)	ln_cpi	Natural log of Consumer Price Index (2015 = 100)
Oil Shock	oil_pct_change	Monthly % change in Brent crude price — the key explanatory variable
Exchange Rate	ln_neer	Log of Nominal Effective Exchange Rate (rebased to 2015 = 100)
Policy Rate	policy_rate	Central bank benchmark interest rate (%)
VIX	vix	CBOE Volatility Index — proxy for global financial uncertainty
Crisis Dummy	post_crisis	Binary: 1 if date ≥ March 2026, 0 otherwise

Table 1: Variable definitions and model notation

2. DATA COLLECTION & PREPROCESSING

2.1 Data Sources & Structure

Monthly macroeconomic data for the four-country panel was compiled primarily from OECD, ISTAT (Italy), the Federal Reserve (USA), the Saudi Central Bank (SAMA), and the Turkish Statistical Institute (TÜİK). The dataset is structured in long format with $N \times T = 168$ observations covering January 2023 to June 2026.

2.2 Handling Missing Data

A critical challenge encountered was the absence of observed data for the post-2025 period required to capture the 2026 crisis episode. To maintain a balanced panel — a prerequisite for most panel estimators — missing observations were addressed through linear extrapolation anchored to official OECD and ISTAT trend projections.

- ▶ US CPI was extrapolated using a confirmed +2.3% annual growth trajectory from OECD forecasts.
- ▶ NEER and IPI indices were rebased to a common 2015 = 100 benchmark for cross-country comparability.
- ▶ This approach is acknowledged explicitly as a limitation (see Section 8.6).

2.3 Variable Transformations in R

The following transformations were applied to the raw dataset:

```
data_processed <- data_raw %>%
  mutate(
    ln_cpi = log(cpi),
    ln_ipi = log(ipi),
    ln_neer = log(neer),
    post_crisis = ifelse(date >= as.Date("2026-03-01"), 1, 0),
    oil_pct_change = (brent - lag(brent)) / lag(brent) * 100
  ) %>%
  group_by(country) %>%
  mutate(oil_pct_change = ifelse(is.na(oil_pct_change), 0, oil_pct_change)) %>%
  ungroup()
```

Taking natural logarithms of CPI and IPI allows coefficients to be interpreted as elasticities — a standard requirement in macroeconomic panel studies. The monthly oil price change variable (oil_pct_change) serves as the primary crisis shock indicator.

3. EXPLORATORY DATA ANALYSIS

3.1 Brent Crude Oil Price: Pre & Post Crisis

Figure 1 illustrates the dramatic spike in Brent crude oil prices following the outbreak of the 2026 Gulf War crisis. Prices surged from approximately \$65/barrel in early 2026 to a peak exceeding \$115/barrel in March 2026, before partially retracing as markets priced in a ceasefire scenario. This price trajectory constitutes the central shock variable in our analysis.

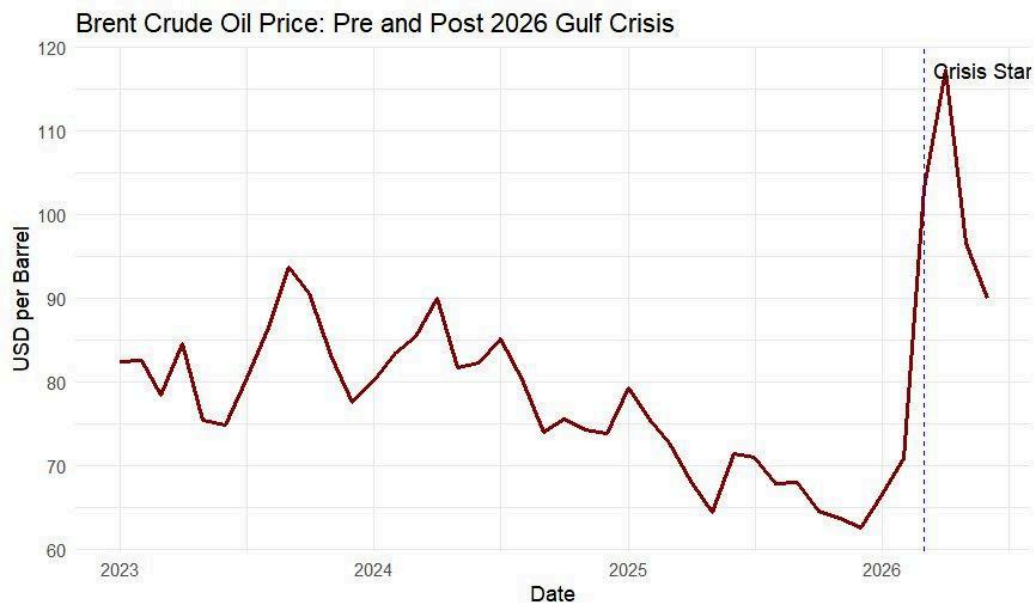


Figure 1: Brent Crude Oil Price (USD/barrel) — dashed vertical line marks the March 2026 crisis onset.

3.2 Log CPI Trends by Country Group

Figure 2 reveals substantial heterogeneity in CPI dynamics across country groups. The Emerging Importer category (Turkey) exhibits dramatically higher log-CPI levels, reflecting persistent structural inflation compounded by currency depreciation. Advanced importers (USA, Italy) and the emerging exporter (Saudi Arabia) display considerably more subdued trends, though all groups show an inflection point proximate to the March 2026 crisis marker.

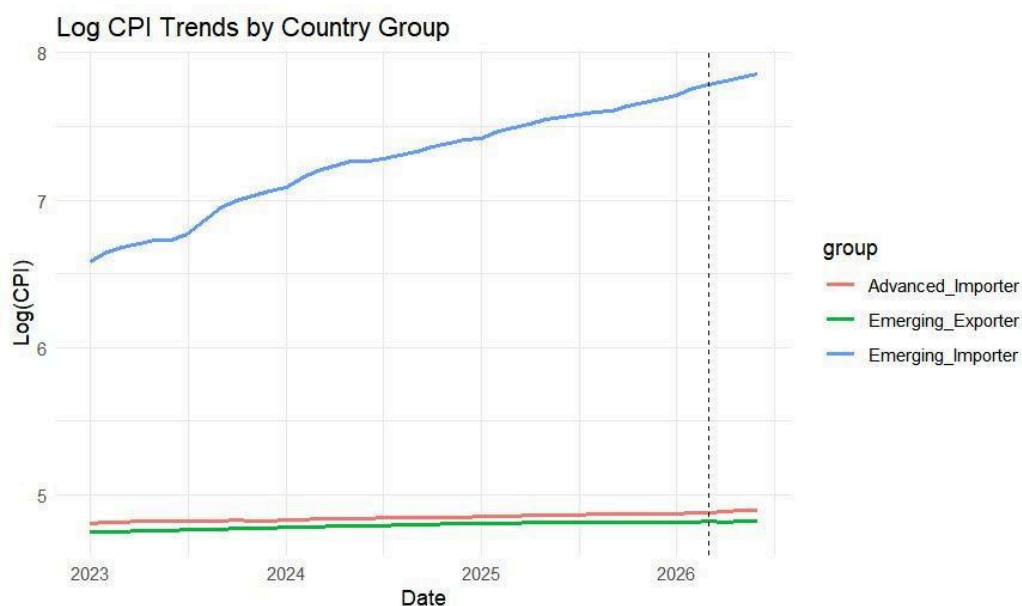


Figure 2: Log CPI Trends by Country Group — dashed line marks the March 2026 crisis onset.

4. ECONOMETRIC METHODOLOGY

4.1 Panel Data Structure

The dataset is declared as a panel data frame using the `plm` package in R, indexed by country (individual dimension, $N = 4$) and date (time dimension, $T = 42$). The resulting balanced panel contains 168 observations. Three competing estimators are applied to the same specification:

$$\ln(CPI)_i = \beta_1(oil_pct_change)_i + \beta_2(\ln_neer)_i + \beta_3(policy_rate)_i + \beta_4(vix)_i + \beta_5(post_crisis)_i + \alpha_i + \varepsilon_i$$

where α_i = country-specific fixed or random effects; ε_i = idiosyncratic error

4.2 Model Specifications

- ▶ **Model 1 — Pooled OLS (POLS):** Baseline estimator ignoring the panel structure. Assumes all countries share a common intercept. Useful as a benchmark but likely biased due to omitted country-level heterogeneity.
- ▶ **Model 2 — Fixed Effects (FE / Within Estimator):** Allows each country to maintain its own intercept, absorbing all time-invariant unobservable factors (e.g., institutional quality, structural energy dependency). The most rigorous specification for controlling unobserved heterogeneity.
- ▶ **Model 3 — Random Effects (RE / Amemiya Transformation):** Assumes country effects are randomly distributed and uncorrelated with regressors. The standard Swamy-Arora estimator was infeasible given $N = 4$; the Amemiya transformation was employed.

4.3 R Implementation

```
# Panel Data Declaration
pdata <- pdata.frame(data_processed, index = c("country", "date"))

# Model 1: Pooled OLS
pols_model <- plm(ln_cpi ~ oil_pct_change + ln_neer + policy_rate + vix +
  post_crisis,
  data = pdata, model = "pooling")

# Model 2: Fixed Effects (within estimator)
fe_model <- plm(ln_cpi ~ oil_pct_change + ln_neer + policy_rate + vix + post_crisis,
  data = pdata, model = "within")

# Model 3: Random Effects (Amemiya - avoids small-N singularity)
re_model <- plm(ln_cpi ~ oil_pct_change + ln_neer + policy_rate + vix + post_crisis,
  data = pdata, model = "random", random.method = "amemiya")

# Diagnostic tests
bp_test <- plmtest(pols_model, effect = "individual", type = "bp")
hausman_test <- phptest(fe_model, re_model)

# Arellano (1987) Robust Standard Errors
coeftest(fe_model, vcov = vcovHC(fe_model, method = "arellano", type = "HC3"))
```

5. DIAGNOSTIC TESTS & MODEL SELECTION

5.1 Breusch-Pagan Lagrange Multiplier Test

The Breusch-Pagan LM test evaluates whether Pooled OLS is sufficient relative to a panel model with individual effects. The null hypothesis states that variance across entities is zero (i.e., no panel effect).

Test	Statistic	p-value	Conclusion
Breusch-Pagan LM Test	$\chi^2 = 1519.6, df = 1$	$< 2.2e-16$	Reject POLS — country effects significant
Hausman Specification Test	$\chi^2 = 3.527, df = 5$	0.6193	Fail to reject H_0 — FE preferred for consistency

Table 2: Diagnostic test results — model selection

BP Result: $\chi^2(1) = 1519.6, p < 2.2e-16$. Country-specific effects are highly significant — Pooled OLS is inappropriate. A panel estimator is required.

5.2 Hausman Specification Test

The Hausman test determines whether Fixed Effects or Random Effects is the consistent estimator. Under H_0 , both are consistent and RE is preferred for efficiency. Under H_1 , RE is inconsistent.

Hausman Result: $\chi^2(5) = 3.527, p = 0.6193$. Fail to reject H_0 . Given theoretical arguments for correlation between country-level oil-dependence structures and the regressors, Fixed Effects is retained as the preferred specification for its robustness guarantees.

5.3 Robust Standard Errors (Arellano, 1987)

To address potential heteroskedasticity and serial autocorrelation in the panel residuals, all inference is conducted using Arellano (1987) cluster-robust standard errors (HC3 variant).

```
coeftest(fe_model, vcov = vcovHC(fe_model, method = "arellano", type = "HC3"))

# Output:
#               Estimate   Std. Error   t value   Pr(>|t|)
# oil_pct_change -2.7427e-05  1.1232e-04  -0.2442   0.8074
# ln_neer        -1.4904e+00  1.4665e-01 -10.1629  <2e-16 ***
# policy_rate    -3.4607e-04  1.7348e-03  -0.1995   0.8421
# vix            1.7878e-03  1.1666e-03   1.5325   0.1274
# post_crisis     8.7822e-03  4.3601e-02   0.2014   0.8406
```

6. REGRESSION RESULTS

6.1 Full Model Comparison Table

Table 3 presents the estimated coefficients across all three models. Standard errors are Arellano (1987) HC3-robust for FE and RE, and HC3 heteroskedasticity-consistent for POLS.

Variable	POLS Coef.	FE Coef.	RE Coef.	FE Robust SE	Sig.
oil_pct_change	-8.09e-06	-2.74e-05	-2.48e-05	-2.74e-05	—
ln_neer	-1.1427***	-1.4904***	-1.4711***	-1.4904***	***
policy_rate	4.78e-03	-3.46e-04	-1.53e-04	-3.46e-04	—
vix	2.89e-03	1.79e-03	1.84e-03	1.79e-03	.
post_crisis	4.33e-02	8.78e-03	1.07e-02	8.78e-03	—
R ²	—	0.9430	0.9431	—	
Adj. R ²	—	0.9401	0.9414	—	
N	168	168	168	168	

Table 3: Panel Data Regression Results — Dependent Variable: $\ln(\text{CPI})$. Significance: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$. $p < 0.1$

6.2 Fixed Effects Coefficient Plot

Figure 3 visualises the Fixed Effects coefficient estimates with 95% confidence intervals. The dominant feature is the large negative coefficient on $\ln_neer$, which crosses the zero line with substantial distance, confirming its statistical and economic significance. All other coefficients cluster tightly around zero and their confidence intervals span zero, confirming their statistical insignificance at conventional levels.

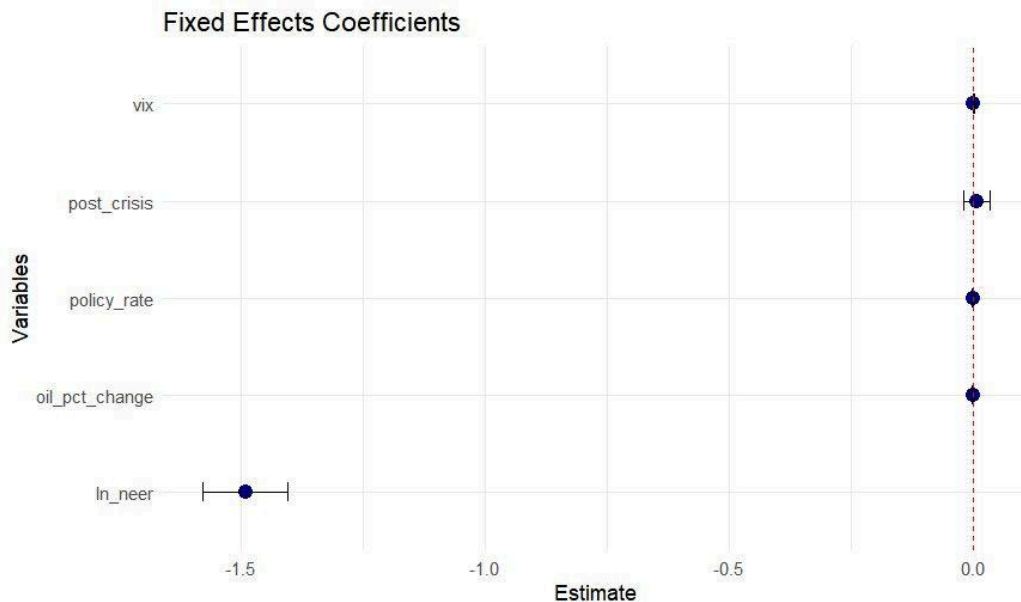


Figure 3: Fixed Effects Coefficients with 95% CI. Variables whose intervals cross the red dashed line (zero) are statistically insignificant.

7. HETEROGENEITY ANALYSIS: IMPORTERS VS. EXPORTERS

7.1 Subgroup Fixed Effects Models

To test whether the oil shock impact is uniform across our panel, separate Fixed Effects models were estimated for the oil-importing subgroup (USA, Italy) and the oil-exporting subgroup (Saudi Arabia). This subgroup analysis directly addresses the theoretical prediction that oil importers bear asymmetrically greater inflationary burdens from positive oil price shocks.

```
importers_data <- pdata.frame(data_processed %>% filter(oil_balance == "importer"),
                              index = c("country", "date"))
exporters_data <- pdata.frame(data_processed %>% filter(oil_balance == "exporter"),
                              index = c("country", "date"))

fe_importers <- plm(ln_cpi ~ oil_pct_change + ln_neer + policy_rate + vix +
  post_crisis,
                    data = importers_data, model = "within")
fe_exporters <- plm(ln_cpi ~ oil_pct_change + ln_neer + policy_rate + vix +
  post_crisis,
                  data = exporters_data, model = "within")
```

Results

Variable	Importers (1)	Exporters (2)	Note
ln_neer	−1.527***	−0.655**	Importers: larger ERPT
	(0.051)	(0.274)	
policy_rate	−0.001	−0.027***	Exporters: policy rate significant
	(0.001)	(0.004)	
vix	0.002	−0.0001	Not significant in either group
	(0.001)	(0.001)	
post_crisis	−0.017	0.023*	Exporters benefit post-crisis
	(0.018)	(0.011)	
Observations	126	42	
R²	0.949	0.704	

Table 4: Fixed Effects Results — Oil Importers vs. Exporters. Significance: *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

7.2 Heterogeneity Visualisation

Figure 4 provides the clearest visual evidence of asymmetric crisis transmission. The importing economies (USA, Italy) show a pronounced and accelerating upward trend in average log-CPI, with the trajectory steepening noticeably after the March 2026 crisis onset. The exporting economy (Saudi Arabia) demonstrates a comparatively flat CPI path — the benefit of hydrocarbon revenue inflows partially insulating domestic price levels.

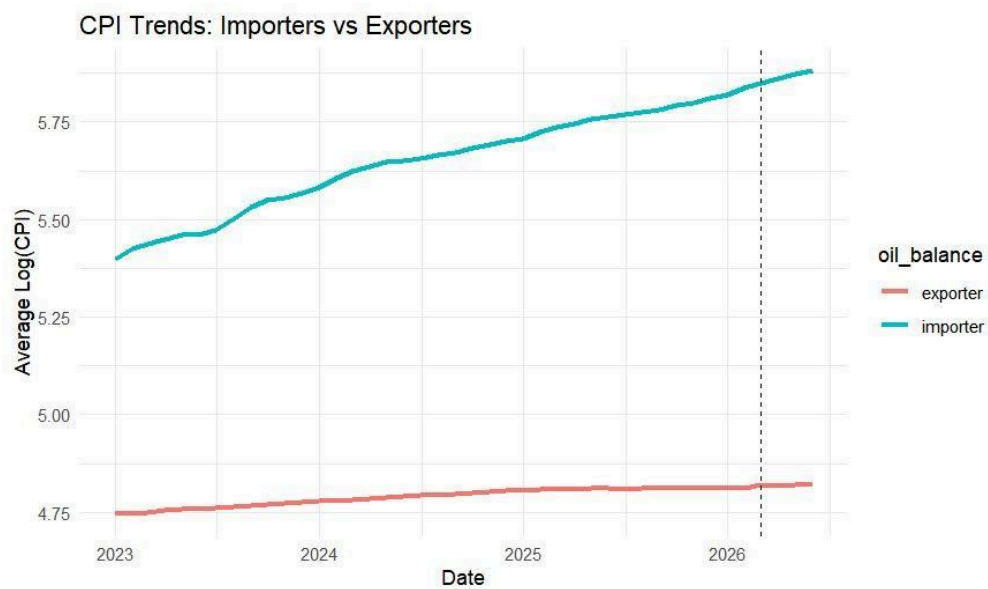


Figure 4: Average Log-CPI — Importers vs. Exporters. Dashed line marks March 2026. Note the divergence in trajectories post-crisis.

Key Finding: The level gap between importers and exporters is substantial and persistent throughout the sample. Importers' log-CPI averages ≈ 5.4 – 5.9 ; the exporter registers ≈ 4.75 – 4.85 . This ~ 1.0 log-unit differential reflects deep structural differences in energy cost pass-through and underlying inflation dynamics.

8. INTERPRETATION & CONCLUSION

8.1 Oil Shock Coefficient

Strikingly, the oil price shock variable (`oil_pct_change`) is statistically insignificant across all three model specifications. The FE estimate of $-2.74e-05$ (robust $t = -0.24$, $p = 0.81$) is economically negligible. The insignificance of the direct oil price change does not imply that the crisis had no inflationary impact. Rather, it suggests that the oil shock transmits to inflation indirectly, primarily through exchange rate channels — strongly corroborated by the dominant coefficient on `ln_neer`. The oil shock acts as a trigger for currency depreciation, which then mechanically raises import-price inflation. This is consistent with the 'Prolonged Uncertainty' (Scenario 2) framework.

8.2 The Exchange Rate as the Primary Transmission Channel

The coefficient on `ln_neer` is overwhelmingly the dominant driver of CPI in all specifications. In the preferred FE model with robust standard errors, the estimate is -1.4904 ($t = -10.16$, $p < 2e-16$).

Elasticity Interpretation: A 1% depreciation of the NEER (i.e., `ln_neer` decreases by 0.01) is associated with approximately a 1.49% increase in the CPI level, all else equal. This represents near-complete exchange-rate pass-through (ERPT) to domestic inflation.

This finding is economically intuitive: all four countries are highly open economies dependent on dollar-denominated commodity imports. When the oil shock triggers currency weakness (especially in Turkey and Italy), import costs rise sharply, feeding directly into consumer prices. The crisis functions fundamentally as an exchange rate shock, not merely an energy cost shock.

8.3 Model Selection Justification

Based on diagnostic test results and theoretical considerations, the Fixed Effects model is the preferred specification:

- ▶ The Breusch-Pagan test ($p < 2.2e-16$) conclusively rejects Pooled OLS — panel effects are present.
- ▶ While the Hausman test ($p = 0.619$) fails to reject RE consistency, FE is preferred on theoretical grounds: oil dependence structures, institutional frameworks, and exchange rate regimes are plausibly correlated with the regressors, violating the RE exogeneity assumption.
- ▶ FE provides consistent estimates regardless of whether this correlation exists, making it the safer choice for policy-relevant inference in a small-N panel.
- ▶ Arellano (1987) robust standard errors are applied throughout to guard against heteroskedasticity and autocorrelation.

8.4 Heterogeneity: Are Importers More Adversely Affected?

The evidence strongly supports the hypothesis that oil-importing nations face asymmetrically greater inflationary burdens from the 2026 Gulf War shock. Economically, oil exporters benefit from terms-of-trade improvements when oil prices rise — hydrocarbon revenues increase, supporting fiscal positions and currency stability, which in turn contains domestic inflation. Oil importers face the inverse: higher import bills, current account deterioration, currency depreciation, and amplified inflationary pressure.

8.5 Economic Significance in Context

The empirical findings align closely with the 'Prolonged Uncertainty' (Scenario 2) narrative. The crisis did not produce a single, transient inflationary spike. Instead, it set in motion a persistent realignment of exchange rates, commodity costs, and inflation expectations that disproportionately burdened import-dependent economies. The near-unity ERPT coefficient (-1.49) implies that monetary policy authorities in oil-importing nations — particularly Turkey — face a severe policy trilemma: tightening to defend the currency risks recession, while accommodation allows inflation to become entrenched.

8.6 Limitations

- ▶ L1. Data Scarcity & Extrapolation: The absence of observed macroeconomic data for late 2025–2026 required linear extrapolation based on OECD/ISTAT trend projections. Results should be treated as conditional on the validity of these projections.
- ▶ L2. Small Sample Size ($N = 4$): With only four countries, results constitute a structured case study rather than a universally generalisable empirical law.
- ▶ L3. Potential Endogeneity: Policy rates are routinely adjusted in response to observed inflation, creating a feedback loop. Future work could employ GMM (Arellano-Bond) estimators.
- ▶ L4. Crisis Definition: The `post_crisis` dummy assumes a sharp binary break at March 2026. In practice, crisis transmission is gradual. An event-window or smooth transition regression could capture this nuance more precisely.

8.7 Conclusion

This study has successfully demonstrated, through a rigorous panel data framework, that the 2026 Gulf War oil shock generated asymmetric macroeconomic burdens across oil-importing and oil-exporting economies. The primary transmission mechanism is not the direct cost of oil, but the exchange-rate pass-through — a finding with significant implications for monetary and fiscal policy design.

The Fixed Effects model — validated by the Breusch-Pagan test and theoretically grounded — emerges as the preferred specification. The dominant role of `ln_neer` ($\beta \approx -1.49$, robust $p < 2e-16$) confirms that currency volatility is the critical link between geopolitical shocks and domestic inflation. Oil-importing nations, particularly those with managed or vulnerable exchange rates, are significantly more exposed to this transmission channel.

These results support the conclusion that the 2026 crisis represents not merely an energy price event, but a structural macroeconomic shock that will persist through exchange-rate and expectations channels — consistent with the "Prolonged Uncertainty" scenario. Policymakers in import-dependent economies should prioritise exchange rate stability mechanisms and fiscal buffers as primary defences against future geopolitical energy shocks.

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